# Seizures

- Measure the blood glucose concentration and treat accordingly.
- Administer midazolam if the seizure is generalised and continues for longer than five minutes, or seizures are recurrent:
  - Administer midazolam IV:
    - 5 mg of midazolam IV for an adult. Consider reducing the dose to 3 mg if a caution is present.
    - See the paediatric drug dose tables for a child.
    - This may be repeated once after five minutes.
  - Administer midazolam IM if IV access has not already been obtained:
    - 10 mg of midazolam IM for an adult. Consider reducing the dose to 5 mg if a caution is present.
    - See the paediatric drug dose tables for a child.
    - This may be repeated once after ten minutes.
- Administer levetiracetam IV if the seizure continues or recurs after two parenteral doses of midazolam:
  - 2 g of levetiracetam for an adult.
  - See the paediatric drug dose tables for a child.
  - Repeat once after ten minutes for an adult, if the seizure continues or recurs ten minutes after the first dose.
- Consider performing RSI if the seizure continues.

## Backup

- Backup from a CCP must be requested if there is severe airway obstruction, persistent severe hypoxia or levetiracetam is indicated.

## Referral and transport

- The patient should receive a clear recommendation to be transported to an ED by ambulance if this is the first time they have had a seizure, unless the cause is clearly recreational drug use and the patient has recovered to be able to mobilise safely.
- The patient may receive a clear recommendation not to be transported to a medical facility by ambulance, even if midazolam has been administered, providing the following criteria are met:
  - No red flags are present.
  - The patient is left in the care of a competent adult.

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- The patient is given a clear recommendation to be seen in primary care within 72 hours for a review of their treatment.

- If transport is required, this should usually be to an ED, but could be to primary care if the patient is rapidly improving and is well known to staff at that facility.

## Red flags (should be seen in ED, usually via ambulance)

- No known history of epilepsy.
- Significant change in their usual pattern of seizures.
- More than one dose of parenteral midazolam administered by ambulance personnel.
- Does not recover to a safe postictal state.
- Injury requiring transport.

## Additional information

### General principles

- Most seizures will terminate spontaneously after 2-3 minutes.
- There is no immediate urgency to treat seizures with medication as long as the patient and their airway are protected by positioning.
- Some patients have pre-prescribed medicines to be administered via the rectal, nasal or buccal route. All personnel may administer such medicines, even if not within their delegated scope of practice, provided they have been prescribed for that patient and the seizure continues for longer than five minutes, or seizures are recurrent.

### Treating seizures with midazolam

- The preferred route is IV, but do not delay administration IM while IV access is obtained because the evidence is that midazolam by the IM route is as effective as IV route.
- Midazolam IV may take 2-3 minutes to stop the seizure.
- A brief period of apnoea is common following successful treatment with midazolam IV.
- The maximum number of doses of midazolam administered by ambulance personnel is two by any route if the patient does not stop seizing. For example, two IV doses, one IM dose and one IV dose, or two IM

doses. This maximum of two doses may be administered in addition to any benzodiazepine that has already been administered prior to ambulance arrival.

- There is no maximum number of midazolam doses if the seizures stop with midazolam administration and the patient has normal/purposeful movement between seizures. Examples of normal/purposeful movement include localising to pain, sitting up, rolling over and pulling on an oxygen mask.
- Wait ten minutes before administering midazolam IV after IM administration. This is to allow adequate time for the IM dose to be absorbed.

## Prolonged seizures

- Prolonged seizures (for example, those lasting greater than 30 minutes) alter brain receptor activity, making pharmacological control of seizures more difficult. The longer the seizure persists before being controlled, the more difficult control becomes.
- Prolonged seizures are life-threatening because they are commonly associated with hypoxia, hypercarbia, metabolic acidosis (lactate is produced during vigorous skeletal muscle activity), hyperthermia (as a result of prolonged muscle activity) and aspiration.

## Following the seizure

- Position the patient on their side.
- Maintain airway and breathing.
- Monitor pulse oximetry and administer oxygen if required.

## The postictal state

- Following a seizure it is common for the patient to have an altered level of consciousness with drowsiness, confusion, agitation or amnesia. This is called the postictal state and usually lasts for 5-60 minutes.
- Sometimes the patient may appear to be in a postictal state, but is actually continuing to have seizures. Suspect this if the patient has rhythmic eye movements, dilated pupils, persistent tachycardia or failure to improve.
- During the postictal state the patient does not usually have capacity to make decisions and it is common for the patient to refuse assessment and/or transport. Clinical judgement is required taking into account the balance of risks, but it may not be in the best interest of the patient to forcibly initiate assessment and/or transport. Seek clinical advice if the situation is difficult to resolve.

# Seizures

## Types of seizures

- Seizures may be classified as generalised (grand mal) or partial (focal or localised). Partial seizures may be simple (where the patient retains awareness of their surroundings) or complex (where the patient loses awareness of their surroundings).
- A patient having partial seizures may present without obvious convulsions or motor activity and may be able to obey commands and interact during seizure activity. The patient may present with any combination of the following:
  - Habitual repetitive movements (automatisms).
  - Sensory symptoms including visual or auditory hallucinations.
  - Emotional outbursts or unusual feelings, such as feeling like they are outside their body.
  - Blank gaze.
- The most common cause of partial seizures is temporal lobe epilepsy.
- Partial seizures may respond to midazolam and/or levetiracetam. If the seizure lasts longer than five minutes and is causing distress, it is appropriate to treat the seizures with anticonvulsants.

## Non-epileptic seizures (pseudo-seizures or psychogenic seizures)

- Non-epileptic seizures occur when there is motor activity that looks clinically like seizures, but there is no electroencephalography (EEG) evidence of seizure activity in the brain.
- In general, healthcare personnel cannot reliably distinguish between epileptic seizures (EEG positive) and non-epileptic seizures (EEG negative).
- It is very common for partial seizures from temporal lobe epilepsy to be initially misdiagnosed as non-epileptic seizures.
- The presence of tongue biting, or incontinence of urine or faeces lowers the likelihood of non-epileptic seizures.
- A patient with non-epileptic seizures may not have conscious control over their motor activity. In the absence of a previously confirmed diagnosis of non-epileptic seizures, treat the patient using the 'seizures' guideline.
- The majority of patients with non-epileptic seizures will subsequently be diagnosed with a medical problem and a proportion have true epilepsy. Only a minority will be diagnosed as having a mental health problem such as conversion disorder or a personality disorder.

# Seizures

## Seizures following recreational drug use

- Seizures following recreational drug use are common, particularly with the chemicals commonly referred to as synthetics.
- The seizures are usually self-limiting and commonly the patient will refuse assessment and/or transport.
- Clinical judgement must be used as forcibly initiating assessment and/or transport often requires a combination of significant restraint and/or sedation, which may be more life-threatening than the effects of the recreational drug.
- It is usually appropriate to manage the patient without transport to an ED if the patient has stopped seizing and has recovered to be able to mobilise safely. Advice on providing supervision until the patient has fully recovered should be provided to family and/or friends whenever feasible.

## Febrile seizures in children

- Febrile seizures are associated with a rapid temperature rise, rather than any specific temperature and usually occur in children aged less than six years.
- The most common cause of febrile seizures is a viral illness.
- Fever associated with infection usually confers some benefit to the patient and does not cause harm provided it is less than 40°C. For this reason rapid and/or aggressive cooling is not indicated unless the temperature is higher than 40°C. Unless this is the case, cool slowly by uncovering the child.

## Seizures during pregnancy

- Eclampsia occurs when a pregnant patient with pre-eclampsia has one or more generalised seizures.
- If the patient is more than 20 weeks pregnant and eclampsia is suspected, refer to the **'eclampsia'** guideline.

## Response and/or transport by helicopter

- Response by helicopter may be indicated if the patient has prolonged seizures and it will arrive at scene at least 15 minutes sooner than a road-based CCP.
- Transport by helicopter may be indicated if the patient has prolonged seizures not resolving with treatment and it will save more than 30 minutes compared to road transport. For criteria to be met, the road transport time will usually need to be more than 90 minutes.

# Seizures

## More Information

### Version history

**16 June 2026** 1.1.0.1

**Update:** Referral and transport guidance updated.

**11 September 2023** 1.0.4

**Update:** Multiple changes.